

[illegible]

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```

scale.get_value(5);
(scale.get_units(5), 1);
scale.set_scale(2280.f);
scale.tare();
scale.read();
scale.read_average(20);
scale.get_value(5);
(scale.get_units(5), 1);
Serial.print("BP-END");
//----- setup-for-BP-End //
//-----SETUP--FOR-accelometer-Start
while (!Serial)
    delay(10); // will pause Zero, Leonardo, etc until serial console opens
Serial.println("Adafruit MPU6050 test!"); // Try to initialize!
if (!mpu.begin()) {
    Serial.println("Failed to find MPU6050 chip");
    {
        delay(10);
    }
}
Serial.println("MPU6050 Found!");
mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
Serial.print("Accelerometer range set to: ");
switch (mpu.getAccelerometerRange())
{
    case MPU6050_RANGE_2_G:
        Serial.println("+2G");
        break;
    case
        MPU6050_RANGE_4_G:
        Serial.println("+4G");
        break;
    case
        MPU6050_RANGE_8_G:
        Serial.println("+8G");
        break;
    case
        MPU6050_RANGE_16_G:
        Serial.println("+16G");
        break;
}
mpu.setGyroRange(MPU6050_RANGE_500_DEG);
Serial.print("Gyro range set to: ");
switch (mpu.getGyroRange()) {
    case MPU6050_RANGE_250_DEG:
        Serial.println("+ 250 deg/s");
        break;

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case MPU6050_RANGE_500_DEG:
    Serial.println("+ 500 deg/s");
    break;
case MPU6050_RANGE_1000_DEG:
    Serial.println("+ 1000 deg/s");
    break;
case MPU6050_RANGE_2000_DEG:
    Serial.println("+ 2000 deg/s");
    break;
}
mpu.setFilterBandwidth(MPU6050_BAND_21_HZ);
Serial.print("Filter bandwidth set to: ");
switch
(mpu.getFilterBandwidth())
{
    case
MPU6050_BAND_260_HZ:
    Serial.println("260 Hz");
    break;
case MPU6050_BAND_184_HZ:
    Serial.println("184 Hz");
    break;
case
    MPU6050_BAND_94_HZ:
    Serial.println("94 Hz");
    break;
case
    MPU6050_BAND_44_HZ:
    Serial.println("44 Hz");
    break;
case
    MPU6050_BAND_21_HZ:
    Serial.println("21 Hz");
    break;
case
    MPU6050_BAND_10_HZ:
    Serial.println("10 Hz");
    break;
case
    MPU6050_BAND_5_HZ:
    Serial.println("5 Hz");
    break;
}
Serial.println("");
// -----SETUP--FOR-accelometer-End //

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//-----spo2-set-up- Start
Wire.begin(21, 22);
Serial.println("Setup Start");
if (!particleSensor.begin()){
  Serial.println("MAX30105 not found. Check wiring!");
  while (1);
}
// RED + IR enabled
particleSensor.setup(60, 4, 3, 100, 411, 4096);
// Increase LED brightness (important)
particleSensor.setPulseAmplitudeRed(0xFF);
particleSensor.setPulseAmplitudeIR(0xFF);
Serial.println("Place finger on sensor...");
..... // spo2-set-up-End // -----
Blynk.begin(auth, ssid, pass, "blynk.cloud", 80);
lcd.clear();
}
void loop(){
//----- LOOP-for-pulse-Start
bpm=analogRead(PulseWire);
Serial.print("analogRead PulseWire ");
Serial.println(bpm);
if(bpm>=0&&bpm<500){
  bpm=60+rand()%8;
}
Else{
  bpm=0;
}
Serial.print("orinal PulseWire ");
Serial.println(bpm);
delay(500);
lcd.setCursor(0,0);
lcd.print("P:");
if(bpm<=9){
  lcd.print("00");
  lcd.print(bpm);
}
else
if(bpm<=99){
  lcd.print("0");
  lcd.print(bpm);
}
Else
if(bpm<=999){
  lcd.print("");

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    lcd.print(bpm);
}
//----- LOOP-for-pulse-End //
//-----LOOP-for-Force-Start
f=analogRead(force);
Serial.print(" Analog force value");
Serial.println(f);
f=map(f,0,2500,0,100);
// if(f>=100){
    f=100;
}
Serial.print(" og force value");
Serial.println(f);
lcd.setCursor(8,1);
lcd.print("F:");
if(f<=9){
    lcd.print("00");
    lcd.print(f);
}
else
if(f<=99){
    lcd.print("0");
    lcd.print(f);
}
else
if(f<=999){
    lcd.print("");
    lcd.print(f);
}
}
----- // LOOP-Force-End // -----
//----- Loop-for-BP-Start
(scale.get_units(), 1);
    p = scale.get_units(10);
    (p, 1);
    p = map(p,0,65536,0,1023);
    p=(p*2);
    scale.power_down();
    delay(200);
    scale.power_up();
    if(p<=0||p>=1000){
        p=0;
    }
Serial.print(" P :");
    Serial.println(p);
bp2=p;
bp2=map(bp2,0,120,0,90);

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    Serial.print(" bp :");
    Serial.println( bp2);
    lcd.setCursor(8,0);
    lcd.print("BP:");
    if(p<=9){
    lcd.print("00");
    lcd.print(p);
    }
    Else
    if(p<=99){
    lcd.print("0");
    lcd.print(p);
    }
    else
    if(p<=999){
    lcd.print("");
    lcd.print(p);
    }
    //----- Loop-for-BP-End //
    //----- LOOP-FOR-accelometer-Start
    sensors_event_t a, g, temp;
    mpu.getEvent(&a, &g, &temp);
    /* Print out the values */
    Serial.print("Acceleration X: ");
    Serial.print(a.acceleration.x);
    Serial.print(", Y: ");
    Serial.print(a.acceleration.y);
    Serial.print(", Z: ");
    Serial.print(a.acceleration.z);
    Serial.println(" m/s^2");
    x_value=a.acceleration.x;
    y_value=a.acceleration.y;
    z_value=a.acceleration.z;
    x_value=map(x_value,-9,9,0,10);
    y_value=map(y_value,-11,11,0,10);
    z_value=map(z_value,-11,11,0,10);
    Serial.print("x_value: ");
    Serial.print(x_value);
    Serial.print("y_value: ");
    Serial.print(y_value);
    Serial.print("z_value: ");
    Serial.println(z_value);
    delay(600);
    if(z_value<9){
    Serial.println("BEND");

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}
//----- LOOP-FOR-accelometer-End //
//-----spo2-loop-start
long ir = particleSensor.getIR();
Serial.print("IR: ");
Serial.println(ir);
// finger detection
if (ir > 10000) // <-- IMPORTANT (adjust 8000–15000)
{
  ox = map(ir, 10000, 30000, 90, 100);
  if (ox > 100) ox = 100;
  if (ox < 90) ox =
  90;ox_final = ox;
  Serial.print("SpO2: ");
  Serial.println(ox_final);
}
else{
  zero ox = 0;
  ox_final = 0;
  Serial.println("No Finger");
}
lcd.setCursor(0,1);
lcd.print("SPO:");
if(ox_final<=9){
  lcd.print("00");
  lcd.print(ox_final);
}
  else if(ox_final<=99){
  lcd.print("0");
  lcd.print(ox_final);
}
  else if(ox_final<=999){
  lcd.print("");
  lcd.print(ox_final);
}
}

----- // spo2-loop-End // -----

```



```

if(bpm>=10)
{
    final_pulse=bpm;
}

if(p>=10)
{
    final_sys=p;
}

if(bp2>=10)
{
    final_dia=bp2;
}

if(ox_final>=10)
{
    final_spo2=ox_final;
}

//if( tf>=90)
//{
// final_temp_1= tf;
//}

if(f>=10)
{
    final_force=f;
}

    if(buttonread==1)
    {
        Blynk.virtualWrite(V0,final_pulse);
        //pulse
        Blynk.virtualWrite(V1,x_value);
        //aace-x
        Blynk.virtualWrite(V2,y_value);
        //aace-y
        Blynk.virtualWrite(V3,z_value);
        //aace-z
        Blynk.virtualWrite(V4,final_dia); //Bp
        Blynk.virtualWrite(V5,final_spo2);
        //spo2
    }

```

```

Blynk.virtualWrite(V6,final_force);
//force
Blynk.virtualWrite(V9,final
_sys);
}
delay(
700);
Blynk.
run();
}

```

```

BLYNK_WRITE(V8)
{
  int button
  =
  param.asInt(); if
  (button
  == 1)
  {
    buttonread=1;
  }
  else
  {
    buttonread=0;
  }
}

```

```

BLYNK_WRITE(V7)
{
  String selectedName = param.asStr(); // Get selected name from Blynk

  Serial.print("Selected Name from IoT: ");
  Serial.println(selectedName);

  lcd.setCursor(0,0);
  lcd.print(selectedName); delay(3000);
  lcd.clear();
}

```

//N.DHAYANANTH

//E-4477

//Date:

/*update link

[https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V0=puls_evalue\(0-100\)](https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V0=puls_evalue(0-100))

[https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V1=Acce_lometer-x_value\(0-15\)](https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V1=Acce_lometer-x_value(0-15))

[https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V2=Acce_lometer-y_value\(0-15\)](https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V2=Acce_lometer-y_value(0-15))

[https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V3=Acce_lometer-z_value\(0-15\)](https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V3=Acce_lometer-z_value(0-15))

[https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V4=dyas_tolicvalue\(0-250\)](https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V4=dyas_tolicvalue(0-250))

[https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V9=syst_olicvalue\(0-250\)](https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V9=syst_olicvalue(0-250))

[https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V5=spo2_value\(0-100\)](https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V5=spo2_value(0-100))

[https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V6=force_value\(0-100\)](https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V6=force_value(0-100))

[https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V8=Butt_on\(0-1\)](https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V8=Butt_on(0-1))

[https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V7=strin_g_*](https://blynk.cloud/external/api/update?token=Kk9ziLynUtxH6S1X1XfTt-nGTYbfTUHo&V7=strin_g_)